INT374 : DATA ANALYTICS WITH POWER BI

PROJECT REPORT

(Project Semester August-December 2026)

Social Media Marketing Analytics



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DECLARATION

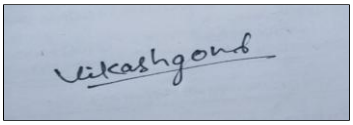
I, Vikash Gond, student of B.Tech Computer Science & Engineering under CSE Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. This project work is original and has not been submitted elsewhere for any academic award.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor Mr. Jaffar Amin Chacket, Assistant Professor at Lovely Professional University, for their continuous support, guidance, and encouragement throughout the course of this project.

I would also like to thank all faculty members of the School of Computer Science and Engineering for providing the resources and learning environment that enabled me to undertake this research. Lastly, I extend my heartfelt thanks to my family and friends for their moral support and motivation.

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1. INTRODUCTION

In today's digital world, social media has become an important platform for communication, brand promotion, customer engagement, and digital marketing. Businesses use social media platforms to reach their target audiences, promote products and services, and understand customer preferences. As the amount of social media data continues to grow, analyzing this information has become essential for evaluating marketing performance and making informed business decisions.

The **Social Media Marketing Analytics** project focuses on analyzing social media performance and marketing campaign data using Microsoft Power BI. The project transforms raw datasets into interactive dashboards that present important performance indicators through charts, graphs, KPI cards, and filters. These visualizations make it easier to understand social media engagement and evaluate the performance of marketing campaigns.

The project consists of two analytical dashboards. The first dashboard, ***Social Media Analytics***, focuses on Instagram performance by examining metrics such as total posts, likes, comments, shares, and engagement rate. It helps users explore content performance and understand how different content categories and media types perform.

The second dashboard, ***Marketing Campaign Analytics***, focuses on marketing campaign performance. It examines indicators such as total campaigns, impressions, clicks, return on investment (ROI), and conversion rate. The dashboard also enables comparisons across campaign types and marketing channels, along with an analysis of impressions and clicks over time.

Microsoft Power BI is used to create interactive visualizations and present the analysis in an organized and understandable format. Data Analysis Expressions (DAX) is used to calculate relevant measures and key performance indicators, while slicers allow users to filter the dashboards and explore different aspects of the datasets.

The primary purpose of this project is to demonstrate how data analytics and visualization can help transform raw social media and marketing campaign data into meaningful information. By presenting key performance indicators and trends in a visual format, the project supports a better understanding of content engagement and marketing campaign effectiveness.

2. OBJECTIVES OF THE PROJECT

2.1 Primary Objective

The primary objective of the project is to develop an interactive Social Media Marketing Analytics dashboard using Microsoft Power BI to analyze social media performance and marketing campaign data.

2.2 Specific Objectives

The specific objectives are:

1. To import and organize social media and marketing campaign datasets in Power BI.
2. To prepare the data for analysis and ensure that relevant fields have appropriate data types.
3. To create calculated measures using Data Analysis Expressions (DAX).
4. To analyze Instagram performance through posts, likes, comments, shares, and engagement rate.
5. To evaluate marketing campaign performance using impressions, clicks, ROI, and conversion rate.
6. To compare campaign performance across campaign types.
7. To compare marketing performance across different channels.
8. To visualize impressions and clicks over time.
9. To create interactive slicers that allow users to filter dashboard information.
10. To present key performance indicators in a structured and understandable format.
11. To develop a report that can be saved and opened in Power BI Desktop.
12. To demonstrate the practical application of data analytics and visualization in digital marketing.

3. PROBLEM STATEMENT

3.1 Problem Definition

Digital marketing activities generate large quantities of data. These records may contain information about social media posts, audience engagement, advertising impressions, clicks, campaign categories, marketing channels, conversion rates, and returns.

When such information is stored in raw spreadsheet or CSV format, understanding overall performance can require considerable manual effort.

Users may find it difficult to compare campaign categories, identify differences between marketing channels, and understand changes in performance over time. Important indicators may also be distributed across multiple columns, making it difficult to obtain a

consolidated view of marketing activities.

The problem addressed by this project is the need for a structured and interactive reporting solution that organizes social media and marketing campaign information into meaningful visualizations.

3.2 Proposed Solution

The proposed solution is to use Microsoft Power BI to build two dashboard pages from the available datasets.

The first page focuses on Instagram social media analytics, while the second focuses on marketing campaign analytics.

The dashboards contain KPI cards, bar charts, column charts, a time-series chart, and interactive slicers. DAX measures are used to calculate important summary values.

This solution allows users to explore the data visually instead of depending exclusively on raw tables.

3.3 Expected Benefits

The proposed solution offers the following benefits:

Improved presentation of marketing information.

Faster access to important performance indicators.

Easier comparison between categories and channels.

Interactive exploration of data.

Reduced dependence on manual calculations.

A consolidated view of important marketing metrics.

Better support for exploratory analysis and reporting.

The dashboard supports decision-making by presenting evidence in a visual format. However, it does not automatically guarantee improved marketing performance or establish the causes of observed trends.

4. SCOPE OF THE PROJECT

4.1 Project Scope

The project covers the analysis and visualization of two datasets using Microsoft Power BI Desktop.

The first dataset is used to analyze Instagram social media performance. The second dataset

is used to examine marketing campaign performance.

The project focuses on descriptive analytics, which involves summarizing historical data and presenting it through charts, graphs, and performance indicators.

4.2 Instagram Analytics Scope

The Instagram dashboard focuses on the following areas:

Total number of posts.

Total likes.

Total comments.

Total shares.

Average engagement rate.

Performance across media types.

Performance across content categories.

Engagement-related comparisons.

Analysis using available date and category filters.

These metrics provide a summarized view of the available Instagram dataset.

4.3 Marketing Campaign Analytics Scope

The marketing campaign dashboard focuses on:

Total campaigns or campaign records, depending on the counting definition.

Total impressions.

Total clicks.

Average ROI.

Average conversion rate.

Impressions by campaign type.

Clicks by campaign type.

Conversion rate by campaign type.

ROI by campaign type.

Campaign count by campaign type.

Impressions by marketing channel.

Clicks by marketing channel.

Conversion rate by marketing channel.

ROI by marketing channel.

Impressions and clicks over time.

The dashboard also includes slicers for campaign type, channel, and date.

4.4 Scope Exclusions

The project does not include predictive modeling, automated advertising optimization, customer-level recommendation systems, or direct integration with live social media platforms.

The dashboards analyze the information contained in the imported datasets. Their conclusions are therefore limited by the accuracy, completeness, and definitions of the available data.

5.TOOLS AND TECHNOLOGIES USED

5.1 Microsoft Power BI Desktop

Microsoft Power BI Desktop is the primary tool used to develop the project.

It provides features for importing data, preparing datasets, creating calculations, building visualizations, and arranging interactive report pages.

Power BI allows multiple visualizations to be displayed together in a report, making it suitable for developing marketing analytics dashboards.

5.2 Data Analysis Expressions (DAX)

DAX is a formula language used in Power BI to create measures and calculated columns.

In this project, DAX measures are used to calculate summary values such as total impressions, total clicks, campaign counts, average ROI, and average conversion rate.

Measures respond to the active filter context, allowing their results to change when users interact with slicers.

5.3 CSV Datasets

Comma-Separated Values (CSV) files are used as the source data for the project.

CSV files store structured information in rows and columns and can be imported into Power BI for further analysis.

The project uses separate datasets for Instagram analytics and marketing campaign analytics.

5.4 Data Visualization

Data visualization converts numerical information into graphical representations.

The project uses:

KPI cards for summary values.

Clustered column charts for comparisons between categories.

Horizontal bar charts for comparisons across channels.

A line chart for examining impressions and clicks over time.

Slicers for interactive filtering.

These visualizations help users explore the datasets from different perspectives.

5.5 DAX Measures Used

The following examples represent measures used or suitable for the project. Field names must match the imported model.

Total Campaign Rows

Campaign Rows Check =

COUNTROWS('marketing_campaigns')

This measure counts the number of rows in the marketing campaign table. It can be used to verify the number of records in the dataset.

Total Impressions

Total Impressions =

SUM('marketing_campaigns'[Impressions])

This measure calculates the sum of the impressions column.

Total Clicks

Total Clicks =

SUM('marketing_campaigns'[Clicks])

This measure calculates the sum of the clicks column.

Average ROI

Average ROI =

AVERAGE('marketing_campaigns'[ROI])

This measure calculates the arithmetic mean of the ROI values.

Average Conversion Rate

Average Conversion Rate =

AVERAGE('marketing_campaigns'[Conversion_Rate])

This measure calculates the arithmetic mean of the conversion rate values.

Instagram Total Posts

Total Posts =

DISTINCTCOUNT('Instagram_Analytics'[post_id])

This measure counts the distinct post identifiers.

Instagram Total Likes

Total Likes =

SUM('Instagram_Analytics'[likes])

This measure calculates the sum of likes.

Instagram Total Comments

Total Comments =

SUM('Instagram_Analytics'[comments])

This measure calculates the sum of comments.

Instagram Total Shares

Total Shares =

SUM('Instagram_Analytics'[shares])

This measure calculates the sum of share

6. DATASET DESCRIPTION

6.1 Dataset Source

The datasets used for this project were obtained from Kaggle, a platform that hosts datasets for data analysis, machine learning, and research.

Dataset source:-

1.Instagram Analytics Dataset: Instagram Analytics — Kaggle

2.Marketing Campaign Performance Dataset: Marketing Campaign Performance — Kaggle

6.2 Instagram Analytics Dataset

The Instagram dataset contains information related to social media accounts, posts, engagement, and content characteristics.

The available fields include identifiers, account information, content categories, media types, likes, comments, shares, impressions, follower-related information, and engagement-related measures.

The following table describes important fields visible in the project data model.

Field	Description
post_id	Identifier used to distinguish posts.
account_id	Identifier for an account.
account_type	Category describing the account type.
likes	Number of likes associated with a record.
comments	Number of comments associated with a record.
shares	Number of shares associated with a record.
impressions	Number of recorded impressions.

engagement_rate	Engagement rate recorded in the dataset.
content_category	Category assigned to content.
media_type	Type of media used in a post.
caption_length	Length of a post caption.
hashtags_count	Number of hashtags associated with content.
follower_count	Follower count recorded for an account or record.
followers_gained	Number of followers gained, as represented in the dataset.
day_of_week	Day associated with the record.

The exact meaning of each field depends on the original dataset's documentation.

The Instagram dashboard uses relevant fields to summarize engagement and compare performance across available categories.

6.3 Marketing Campaign Dataset

The marketing campaign dataset contains information about digital marketing campaigns, their channels, and selected performance indicators.

Important fields available in the Power BI data model include:

Field	Description
Campaign_ID	Identifier associated with a campaign record.
Campaign_Type	Category of marketing campaign.
Channel_Used	Marketing channel used.
Impressions	Number of recorded impressions.
Clicks	Number of recorded clicks.
Conversion_Rate	Conversion rate recorded for a campaign record.
ROI	Return-on-investment value recorded in the dataset.
Date	Date associated with the campaign record.
Company	Company associated with the record.

Acquisition_Cost	Acquisition cost recorded in the dataset, where applicable.
Customer_Segment	Customer segment associated with the record.
Duration	Duration information available in the dataset.
Engagement_Score	Engagement-related score recorded in the dataset.
Language	Language field associated with the record.
Location	Location field associated with the record.
Target_Audience	Target audience category.

The table may contain additional fields. The final description should reflect the columns present in the submitted dataset.

6.4 Data Characteristics

The two datasets serve different analytical purposes.

The Instagram dataset focuses on content performance and audience engagement. The marketing campaign dataset focuses on advertising and campaign-related performance metrics.

Although both datasets relate to marketing, they do not share an established common identifier suitable for joining their records. Consequently, they are analyzed independently on separate report pages.

This approach avoids creating an unsupported relationship between unrelated records.

6.5 Data Quality Considerations

- Before interpreting results, it is important to consider:
- Whether records contain missing values.
- Whether identifiers are unique or repeated.
- Whether numerical fields have appropriate data types.
- Whether dates are correctly interpreted.
- Whether conversion rates are stored as decimals or percentages.
- Whether ROI represents a ratio, percentage, or another numerical definition.
- Whether the aggregation used for each metric is appropriate.

These checks help ensure that visualizations are interpreted correctly.

7. METHODOLOGY

7.1 Overview

The project follows a sequence of steps to transform raw datasets into interactive dashboards.

The main stages are:

1. Dataset collection.
2. Data import.
3. Data inspection and preparation.
4. Measure creation.
5. Dashboard development.
6. Interactive filter implementation.
7. Testing and validation.
8. Report preparation.

7.2 Dataset Collection

The datasets were obtained from Kaggle and prepared for import into Microsoft Power BI Desktop.

Separate datasets were selected to examine Instagram social media performance and digital marketing campaign performance.

7.3 Data Import

The CSV files were imported into Power BI Desktop.

After import, the tables were reviewed in the Data and Model views to identify available columns, data types, and the structure of the datasets.

The tables used in the project are:

- Instagram_Analytics
- marketing_campaigns

7.4 Data Inspection and Preparation

The imported datasets were inspected to understand their fields and determine which columns were appropriate for analysis.

Relevant fields were used to create measures and visualizations.

Date, categorical, and numerical columns were used according to their analytical purpose.

Data preparation is important because incorrect data types or inappropriate aggregation settings can produce misleading results.

7.5 Measure Creation

DAX measures were created to summarize important values.

Examples include total impressions, total clicks, average ROI, average conversion rate, total posts, total likes, and total comments.

These measures allow Power BI to recalculate summary values according to the filters applied by users.

7.6 Dashboard Development

Two report pages were developed within a single Power BI file.

The first page presents Instagram analytics. The second presents marketing campaign analytics.

Each page contains a title, KPI cards, charts, and relevant slicers.

The visualizations are arranged to make key information accessible without requiring users to inspect the underlying tables.

7.7 Interactive Analysis

Slicers were added to allow users to filter the report.

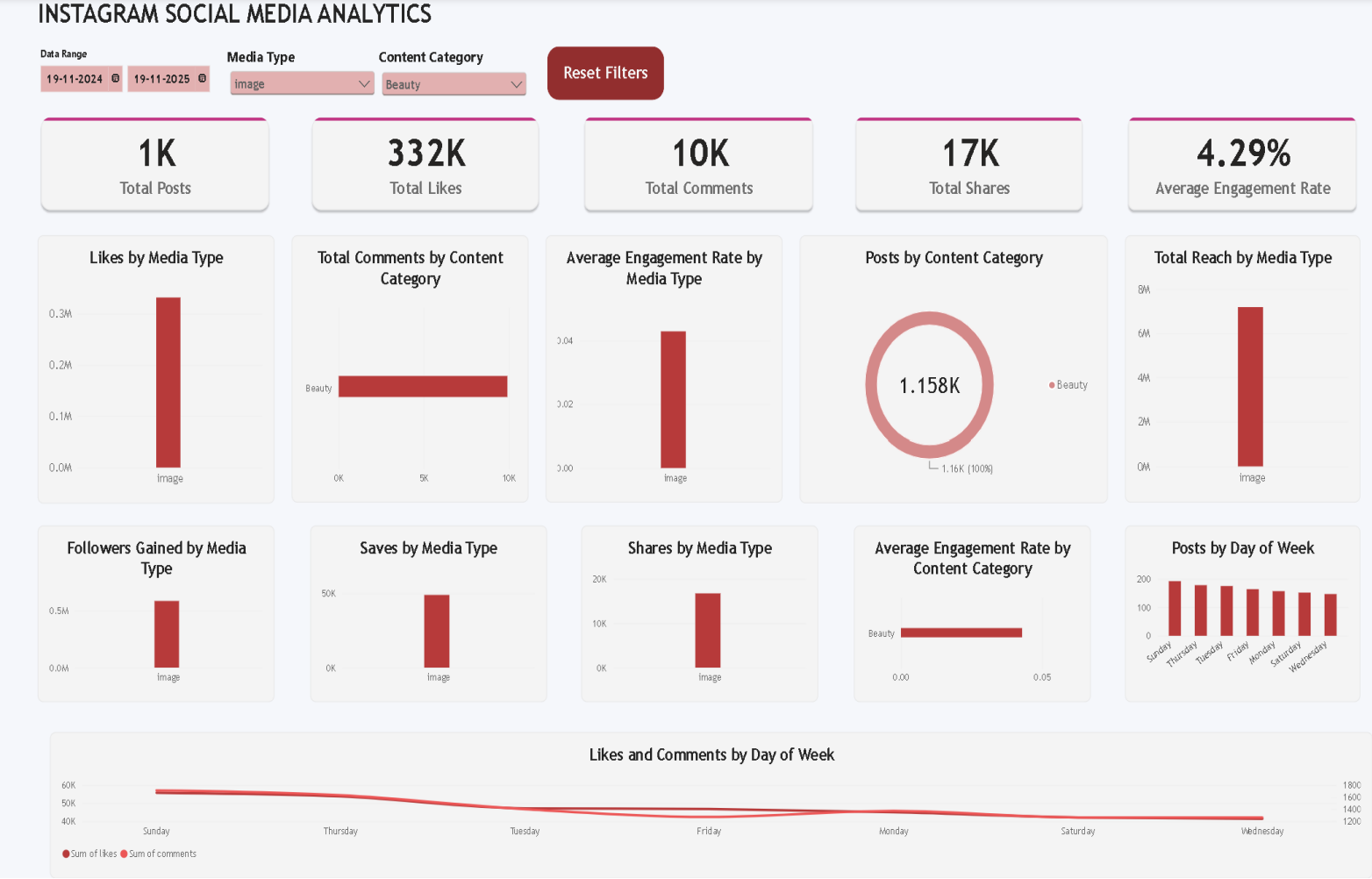
The marketing campaign page includes campaign type, channel, and date filters. The Instagram page uses relevant date and content-related filters.

When a slicer selection changes, the connected visualizations update according to the selected filter context.

7.8 Testing and Validation

The completed report was reviewed to check that the visualizations rendered, slicers responded, and both report pages remained accessible.

Numerical results should also be compared against the underlying data to confirm that measures and aggregations are correct.



8. SYSTEM DESIGN AND ARCHITECTURE

8.1 System Architecture

The system follows a straightforward data-analysis architecture.

Data Sources → Data Import → Data Preparation → DAX Measures → Visualizations → Interactive Dashboards

Stage 1: Data Sources

The system uses the Instagram analytics dataset and the marketing campaign dataset.

Stage 2: Data Import

The CSV files are imported into Power BI Desktop as separate tables.

Stage 3: Data Preparation

The fields are inspected and prepared for use in calculations and visualizations.

Stage 4: Calculation Layer

DAX measures calculate summary values from the imported tables.

Stage 5: Visualization Layer

KPI cards, bar charts, column charts, and line charts display the calculated information.

Stage 6: Interaction Layer

Slicers allow users to filter information by selected categories and dates.

Stage 7: Report Output

The finished report is saved as a Power BI Desktop file with the extension .pbix.

8.2 Data Model

The Power BI model contains two separate tables:

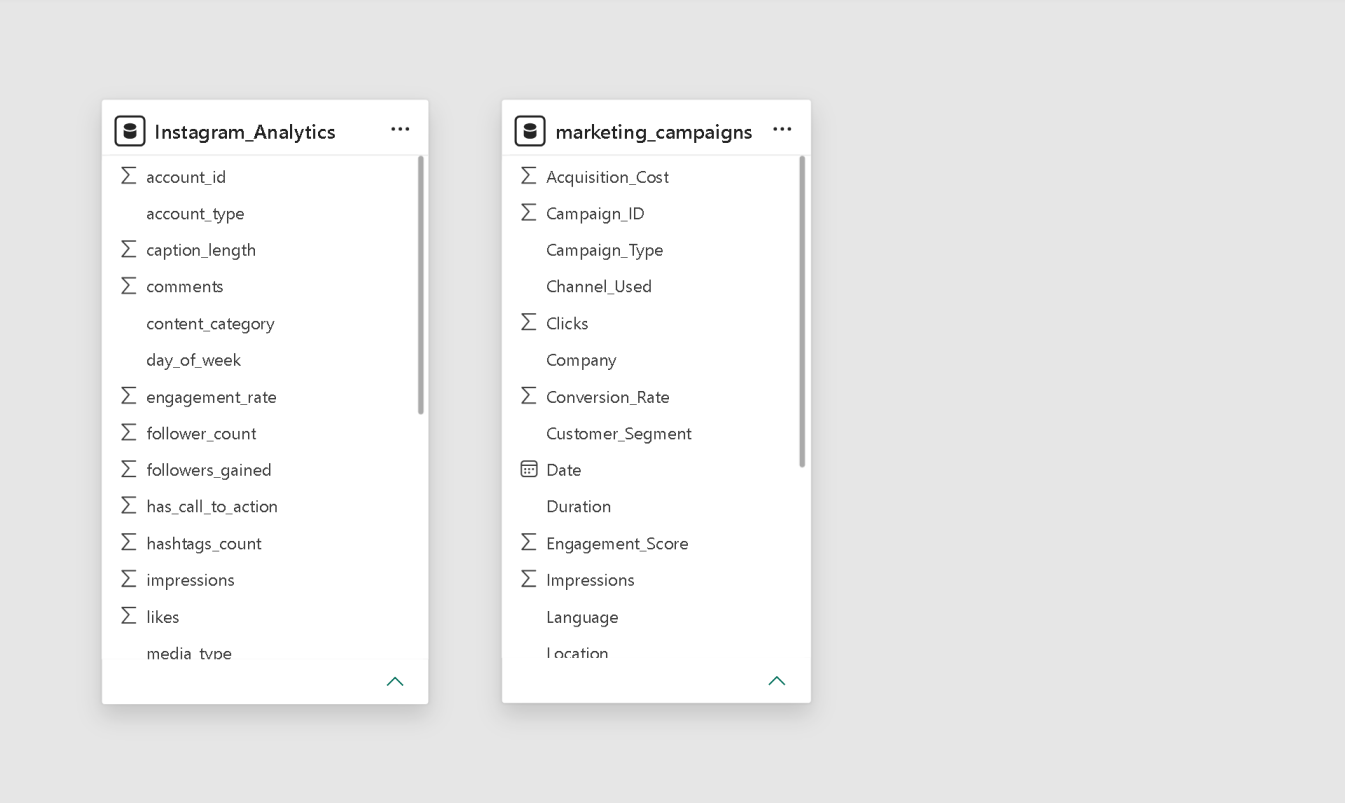
Instagram_Analytics

marketing_campaigns

The tables represent different datasets and do not have an established common key for a meaningful relationship.

Therefore, the dashboards analyze each table independently.

This design keeps the two analytical areas separate and avoids unsupported joins.



8.3 Report Structure

The report contains two primary pages.

Page 1: Social Media Analytics

This page focuses on Instagram performance and engagement.

Page 2: Marketing Campaign Analytics

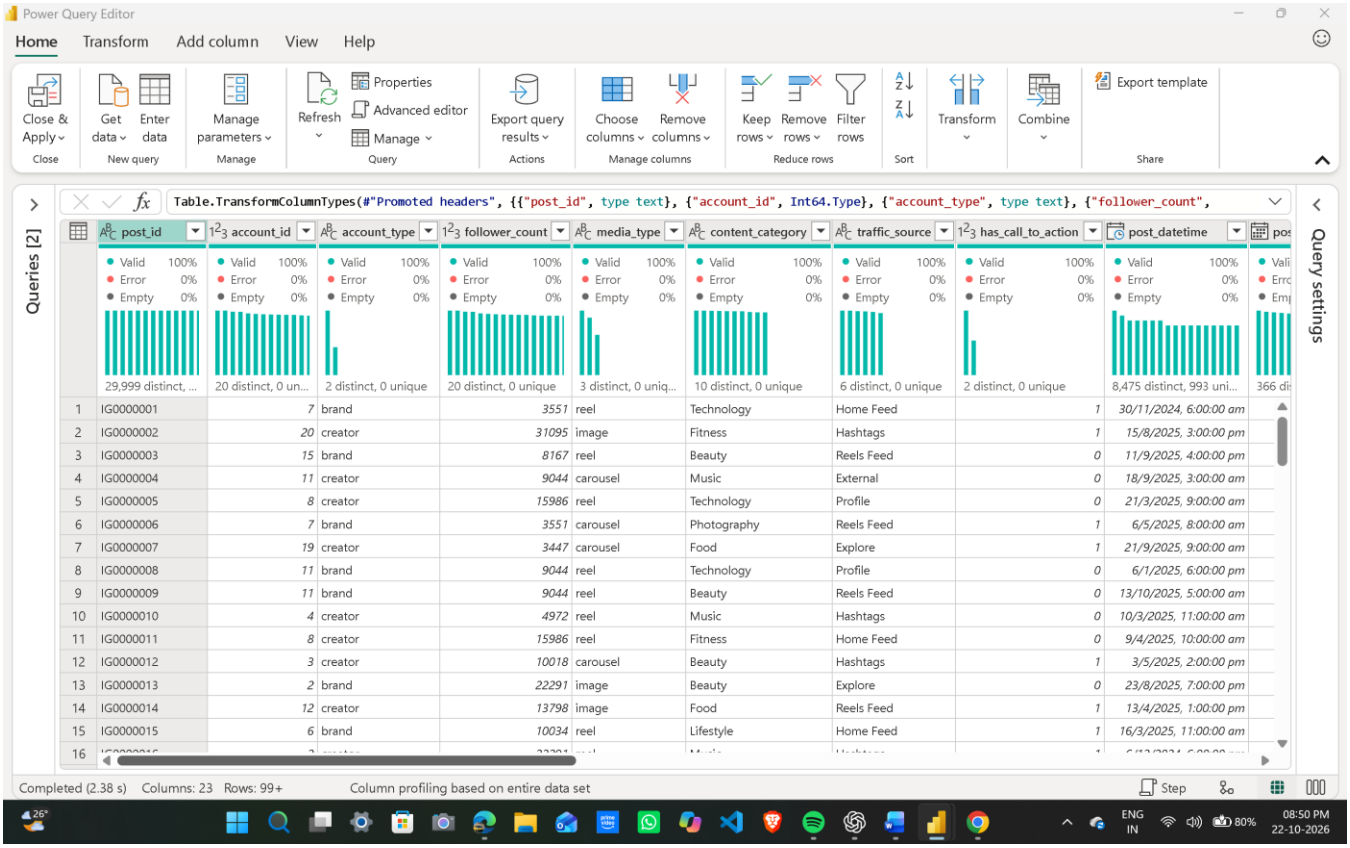
This page focuses on campaign performance, channel comparisons, and time-based analysis.

Users can switch between the two pages using the report page tabs.

9. DATA PREPARATION AND TRANSFORMATION

9.1 Importance of Data Preparation

Data preparation helps ensure that fields can be used correctly in calculations and visualizations.
For example, a date column must be interpreted as a date to support time-based analysis. Numerical fields must use appropriate numeric data types to support summation or averaging.
Incorrect preparation can cause errors in charts, unexpected aggregations, or misleading results.



9.2 Column Selection

Relevant columns were selected according to the purpose of each dashboard.
For Instagram analysis, the important fields include post identifiers, likes, comments, shares, engagement rate, content category, media type, and date-related information.
For campaign analysis, the important fields include campaign type, channel, impressions, clicks, conversion rate, ROI, campaign identifier, and date.

9.3 Data Types

The relevant columns should use suitable data types.

Data Category	Suitable Data Type
Identifiers	Text or whole number, depending on the source.
Likes and comments	Whole number.
Shares	Whole number.
Impressions	Whole number.
Clicks	Whole number.
Conversion rate	Decimal number or percentage, according to the source.
ROI	Decimal number or percentage, according to the source definition.
Date	Date or date/time.
Campaign type	Text.
Channel used	Text.
Content category	Text.
Media type	Text.

Identifiers should not automatically be summed merely because they contain numbers.

9.4 Aggregation Methods

Different metrics require different aggregation methods.

- **SUM:** Appropriate for additive metrics such as clicks, impressions, likes, comments, and shares, provided the records and metric definitions support summation.
- **AVERAGE:** Appropriate when the goal is to calculate the arithmetic mean of individual conversion-rate or ROI values.
- **DISTINCTCOUNT:** Appropriate when counting unique post or campaign identifiers.
- **COUNTROWS:** Appropriate when counting records in a table.
-

The correct method depends on the meaning of the field and the structure of the dataset. For example, counting campaign records is not necessarily the same as counting unique campaigns.

9.5 Handling Conversion Rate

Conversion rates may be stored as decimals or as percentage values.
For example, a stored value of 0.08 may represent 8%, while a stored value of 8 may already represent 8%.
Therefore, percentage formatting must match the underlying data representation.
The measure should not be multiplied by 100 without checking how the source data stores the values.

9.6 Data Model Separation

The two datasets were retained as separate analytical tables because they do not have a suitable common identifier.
This allows each dashboard to use the correct source fields and calculations without mixing unrelated records.

10.DASHBOARD DEVELOPMENT

10.1 Overview

The report was developed in Microsoft Power BI Desktop using a consistent visual layout.

Both pages use KPI cards, chart titles, and coordinated formatting to present the information in a structured manner.

The report uses contrasting text and chart colors against a light background to make the visualizations easier to distinguish.

10.2 Dashboard 1: Social Media Analytics

Purpose

The Social Media Analytics dashboard provides a summarized view of Instagram performance.

It is designed to help users understand post activity and engagement through visual indicators and category-based comparisons.

KPI Cards

The dashboard includes the following main indicators:

Total Posts

Total Likes

Total Comments

Total Shares

Average Engagement Rate

These indicators provide a high-level summary of the selected data.

Visualizations

The dashboard includes visualizations that examine different aspects of Instagram performance.

Examples of analytical views include:

Post performance comparisons.

Likes and comments analysis.

Shares analysis.

Engagement-rate analysis.

Comparisons by media type.

Comparisons by content category.

Date-based analysis.

The purpose of these visualizations is to help users explore relationships between content characteristics and engagement-related metrics.

Slicers

Slicers allow users to filter the dashboard using available fields.

The dashboard includes date and content-related filtering options, including media type and content category.

Users can select a category to examine the corresponding data in the connected visualizations.

Expected Use

A user can examine the overall KPI cards and then select a media type or content category to explore how the associated metrics change.

This makes it possible to investigate the performance of specific content groups without

manually filtering the raw dataset.

10.3 Dashboard 2: Marketing Campaign Analytics

The Marketing Campaign Analytics dashboard presents a summarized view of marketing campaign performance.

It allows users to compare campaign categories and marketing channels using impressions, clicks, ROI, and conversion rate.

KPI Cards

The dashboard contains five primary KPI cards.

KPI	Purpose
Total Campaigns	Presents the campaign-count measure used in the report.
Total Impressions	Summarizes recorded impressions.
Total Clicks	Summarizes recorded clicks.
Average ROI	Displays the average of the ROI field.
Average Conversion Rate	Displays the average conversion rate.

The campaign-count measure should be clearly defined in the final report. A distinct count of Campaign_ID represents unique identifiers, whereas COUNTROWS represents the number of records.

Campaign Type Visualizations

The dashboard includes the following campaign-type comparisons.

1. Impressions by Campaign Type

This column chart compares the sum of impressions across campaign types.

It provides a visual overview of recorded impressions for the available categories.

2. Clicks by Campaign Type

This chart compares the sum of clicks across campaign types.

It helps users examine differences in click volume between categories.

3. Conversion Rate by Campaign Type

This chart compares the average conversion rate across campaign types.

It provides a view of how conversion-rate values differ among the available categories.

4. ROI by Campaign Type

This chart compares average ROI across campaign types.

It allows users to examine differences in the recorded ROI values.

5. Campaign Count by Campaign Type

This horizontal bar chart presents the campaign-count measure by campaign type.

The interpretation depends on whether the measure counts unique campaign identifiers or rows in the dataset.

Channel Performance Visualizations

The dashboard also contains comparisons across marketing channels.

6. Impressions by Channel

This chart compares total impressions across the available marketing channels.

7. Clicks by Channel

This chart compares total clicks across marketing channels.

8. Conversion Rate by Channel

This chart compares average conversion rate across channels.

9. ROI by Channel

This chart compares average ROI across marketing channels.

Together, these visualizations provide a way to examine channel-level differences in the recorded campaign metrics.

Time-Series Visualization

10. Impressions and Clicks Over Time

The dashboard includes a line chart displaying impressions and clicks across the available dates.

This visualization helps users explore how the two metrics change over time.

Because impressions and clicks may differ substantially in magnitude, their axis scales should be reviewed carefully. A dual-axis or separate-chart approach may improve interpretability when necessary.

Slicers

The Marketing Campaign Analytics page includes three slicers:

Campaign_Type

Channel_Used

Date

The campaign-type slicer filters the report according to the selected campaign category.

The channel slicer allows users to focus on a specific marketing channel.

The date slicer allows users to examine the data for a selected date or date range.

These filters make the dashboard interactive and allow users to explore subsets of the dataset.

10.4 Visual Design

The dashboards use a consistent design approach.

The main design features include:

Clear page titles.

KPI cards arranged near the top.

Charts arranged in a structured layout.

Light backgrounds.

Coordinated chart colors.

Readable chart titles.

Consistent spacing between visualizations.

Interactive slicers positioned near the top of the report pages.

The purpose of the design is to provide a consistent and understandable reporting experience.

11.KEY PERFORMANCE INDICATORS

11.1 Meaning of KPIs

Key Performance Indicators are measurable values used to summarize performance against a specific objective.
In this project, KPIs provide an overview of social media engagement and marketing campaign activity.
Their interpretation depends on the definitions and quality of the underlying data.

11.2 Total Posts

Total Posts measures the number of distinct post identifiers in the Instagram dataset.
It provides an overview of the post records represented in the report.
If post identifiers repeat across records, a distinct count is more appropriate than a simple row count when measuring unique posts.

11.3 Total Likes

Total Likes summarizes the likes recorded in the Instagram dataset.
It provides a measure of accumulated engagement through likes.
The result should be interpreted according to how likes are represented in the source data.

11.4 Total Comments

Total Comments summarizes the recorded comments.
Comments can be used as an engagement indicator, although the total alone does not reveal whether comments are positive, negative, or neutral.

11.5 Total Shares

Total Shares summarizes the recorded shares.
Shares can provide information about how frequently content is redistributed, provided the dataset's definition of a share is consistent.

11.6 Average Engagement Rate

Average Engagement Rate summarizes the engagement-rate values in the Instagram dataset.
The calculation used in the project is an arithmetic average of the recorded values.
If engagement rates are calculated from different denominators or groups of different sizes, the simple average may differ from an overall weighted engagement rate.

11.7 Total Impressions
Total Impressions calculates the sum of recorded impressions in the marketing campaign dataset.
Impressions represent recorded displays of content or advertisements, according to the source dataset's definition.
Impressions do not necessarily represent unique people, because a person may generate multiple impressions.

11.8 Total Clicks
Total Clicks calculates the sum of the clicks recorded in the marketing campaign dataset.
Clicks provide a measure of interaction with the relevant marketing content.
The number of clicks alone does not indicate whether those interactions resulted in conversions.

11.9 Average ROI
Average ROI calculates the arithmetic mean of the values in the ROI column.
ROI is commonly used to evaluate returns relative to investment. However, the exact formula and scale used in this dataset must be checked before interpreting the displayed value.
A simple average of campaign-level ROI values is not necessarily equivalent to overall ROI calculated from total returns and total investment.

11.10 Average Conversion Rate
Average Conversion Rate calculates the arithmetic mean of the recorded conversion-rate values.
It provides a summary of the conversion-rate values represented in the dataset.
The value should be formatted and interpreted according to the source's storage convention.

12. DATA ANALYSIS AND INTERPRETATION

12.1 Introduction
The dashboards provide a visual environment for exploring Instagram performance and marketing campaign activity.
The analysis focuses on summarizing the available data, comparing categories, and examining changes over time.
The dashboards are descriptive in nature. They show patterns in the dataset but do not, by themselves, establish why those patterns occur.

12.2 Instagram Performance Analysis

The Instagram dashboard presents summary indicators related to posts, likes, comments, shares, and engagement rate.
These indicators provide an overview of recorded content activity.
Comparisons across content categories and media types allow users to explore whether the recorded engagement metrics differ between groups.
Date-based filtering can also help users examine specific periods.
However, the presence of a difference between categories does not establish that the content category or media type caused the difference. Other factors, such as audience size, posting frequency, and the characteristics of the content, may influence the results.

12.3 Campaign Type Analysis

The marketing campaign dashboard compares impressions, clicks, conversion rate, and ROI across campaign types.
Impressions by campaign type provide a view of recorded exposure.
Clicks by campaign type provide a view of interaction volume.
Conversion rate by campaign type helps compare the recorded conversion-rate values, while ROI by campaign type provides a comparison of the recorded return-related values.
The campaign-count chart adds information about how campaign records or unique campaign identifiers are distributed across categories.
These views can be used to identify categories for further investigation.

12.4 Marketing Channel Analysis

The channel-level charts allow users to compare recorded impressions and clicks across marketing channels.
The conversion-rate and ROI charts provide additional perspectives beyond activity volume.
For example, a channel may have a large number of impressions but a different average conversion rate from another channel. Such a difference can be explored through the dashboard, but further investigation is required before concluding which channel performs better overall.
A meaningful evaluation should consider the metric definitions, campaign objectives, costs, and other relevant information.

12.5 Time-Based Analysis

The line chart showing impressions and clicks over time allows users to inspect changes across the available dates.
This view can help identify periods with unusually high or low values and possible recurring patterns.
However, apparent peaks or fluctuations should be checked against the underlying data

before being interpreted as meaningful events.
The time-series visualization is a descriptive tool rather than a forecasting model.

12.6 Interactive Analysis

The slicers allow users to investigate different subsets of the data.
For example, a user can select a campaign type and examine the associated KPI values and chart results.
The channel slicer can be used to focus on a particular marketing channel, while the date slicer can restrict the analysis to a selected period.
The interaction between filters and visualizations helps users explore the dataset without creating separate reports for every category.

12.7 Interpretation of Results

The main analytical value of the project is its ability to organize multiple performance indicators into an interactive report.
The dashboards make it possible to compare categories, inspect recorded activity, and examine relationships between metrics.
The actual conclusions about the best-performing campaign type, channel, or content category must be based on verified values from the final Power BI report.
Before reporting a category as the best performer, the relevant metric should be checked, the correct aggregation should be confirmed, and the effect of the selected filters should be considered.

13. TESTING AND VALIDATION

13.1 Purpose of Testing

Testing is important to ensure that the report displays data correctly and that users can interact with the dashboards as intended.
The project was reviewed in Power BI Desktop to examine its pages, visualizations, and slicers.

13.2 Functional Testing

The following table describes the principal functional tests for the report.

Test ID	Test Description	Expected Result
T01	Open the Power BI report.	Both report pages are accessible.
T02	Inspect Instagram KPI cards.	The cards display values calculated from the Instagram dataset.
T03	Inspect marketing KPI cards.	The cards display values calculated from the campaign dataset.

T04	Select a campaign type.	Connected campaign visualizations respond to the filter.
T05	Select a marketing channel.	Connected campaign visualizations respond to the filter.
T06	Change the date range.	Date-sensitive visualizations update appropriately.
T07	Compare campaign-type charts.	The charts use the intended fields and aggregation methods.
T08	Inspect the time-series chart.	Dates and metrics are displayed in the intended visual.
T09	Switch between report pages.	Each page opens and displays its corresponding dashboard.
T10	Save and reopen the PBIX file.	The report opens with its saved structure and visuals.

The table describes the validation checklist. Each test should be marked as passed only after its result has been verified.

13.3 Numerical Validation

Numerical validation should be performed by comparing the results of key measures against the underlying data.

For example:

- Total impressions should be compared with the sum of the Impressions column.
- Total clicks should be compared with the sum of the Clicks column.
- Campaign counts should be checked against the distinct campaign identifiers or row count, depending on the measure definition.
- Average ROI should be checked against the average of the ROI values.
- Average conversion rate should be checked against the average of the conversion-rate values.

This helps identify errors caused by incorrect aggregation, unexpected duplicate records, or unsuitable data types.

13.4 Visual Validation

Visual validation involves checking:

- Chart titles.
- Axis labels.
- Category labels.
- KPI card formatting.
- Percentage formatting.
- Chart readability.

- Slicer behavior.
- Visual alignment.
- Report page layout.

These checks help ensure that the report is understandable and suitable for presentation.

13.5 Aggregation Validation

The report uses different aggregation methods depending on the metric.
Additive metrics, such as impressions and clicks, are generally summarized using SUM.
ROI and conversion rate are summarized using AVERAGE in the relevant charts and measures.
Campaign counts require special attention because counting rows and counting unique identifiers produce different results when identifiers repeat.
The final report should use the measure that matches the intended definition of campaign count.

14. RESULTS AND DISCUSSION

14.1 Project Results

The project produced a Power BI report containing two analytical dashboard pages.
The first page presents Instagram social media performance. The second page presents marketing campaign performance.
The report brings together KPI cards, categorical comparisons, channel analysis, date filtering, and a time-series visualization.

14.2 Instagram Dashboard Results

The Instagram dashboard organizes key engagement-related indicators into a single report page.
It allows users to review the selected metrics and explore performance across available content categories and media types.
The use of slicers enables users to investigate subsets of the data without manually editing the source file.

14.3 Marketing Dashboard Results

The Marketing Campaign Analytics dashboard summarizes impressions, clicks, average ROI, and average conversion rate.
It also presents comparisons by campaign type and marketing channel.
The line chart provides a time-based view of impressions and clicks, while the campaign type, channel, and date slicers support interactive analysis.

14.4 Benefits of the Developed Report

The developed report offers several practical benefits:

1. **Centralized reporting:** Multiple metrics are presented together in a structured dashboard.
2. **Visual comparison:** Charts make it easier to compare categories and channels.
3. **Interactive filtering:** Users can explore selected categories and periods.
4. **Consistent calculations:** DAX measures provide reusable calculations across the report.
5. **Improved accessibility:** KPI cards provide a quick overview of selected metrics.
6. **Separate analytical perspectives:** Instagram performance and campaign performance are organized into different pages.
7. **Reusable report format:** The PBIX file can be reopened in Power BI Desktop for further analysis and modification.

14.5 Discussion

The project demonstrates the practical use of business intelligence tools for marketing analytics.

It shows how data can be summarized and presented in a form that is easier to explore than raw CSV tables.

The value of the report depends on the accuracy of the imported data, the suitability of the DAX calculations, and the interpretation of the visualized results.

The dashboards support descriptive analysis, but further work would be required to establish causal relationships, predict future performance, or recommend optimal marketing strategies automatically.

15.LIMITATIONS

Although the project provides an interactive environment for exploring marketing data, several limitations should be considered.

15.1 Dependence on Available Data

The analysis is limited to the fields and records available in the imported datasets.

Missing, inaccurate, or inconsistent records may affect the calculated metrics and the interpretation of the results.

15.2 Historical Data

The dashboards analyze the imported data rather than automatically retrieving live information from social media platforms or advertising systems.

Consequently, the report reflects the available dataset rather than necessarily representing current marketing performance.

15.3 Separate Datasets
The Instagram and marketing campaign datasets are analyzed independently because no suitable shared key has been established for joining their records.
The report therefore does not provide a unified record-level analysis connecting individual Instagram posts with individual marketing campaigns.

15.4 Aggregation Limitations
A simple average of ROI or conversion-rate values may not represent the overall rate across all records.
Similarly, counting rows does not necessarily measure unique campaigns.
The interpretation of these metrics depends on the source definitions and the aggregation method used.

15.5 Descriptive Analytics
The project focuses on descriptive analytics rather than prediction or optimization.
It summarizes the available data but does not automatically forecast future results or determine the causes of performance differences.

15.6 Visual Interpretation
Charts can make comparisons easier, but they may also be misleading if axes, scales, aggregation methods, or formatting are inappropriate.
The report should therefore be reviewed to ensure that the visualizations communicate the data accurately.

16. FUTURE ENHANCEMENTS
The project can be extended in several ways.

16.1 Additional KPIs
Additional indicators could be introduced if the necessary fields and definitions are available.
Examples include click-through rate, cost per click, cost per acquisition, and weighted conversion rate.
These measures should only be implemented when the required source data is available and the formulas are appropriate.

16.2 Improved Data Validation
Additional checks could be introduced to identify duplicate identifiers, missing values, inconsistent dates, and unexpected numerical values.
This would improve confidence in the calculations and visualizations.

16.3 Improved Campaign Evaluation

The campaign dashboard could be expanded to compare campaign performance while considering investment, costs, and business outcomes.
This would provide a more comprehensive basis for evaluating marketing effectiveness than relying on impressions or clicks alone.

16.4 Advanced Time-Series Analysis

The existing time-series visualization could be expanded with monthly summaries, moving averages, or other techniques for examining temporal patterns.
Forecasting could be added as a separate extension if suitable historical data is available.

16.5 Additional Filters

More filters could be introduced based on the availability and relevance of fields such as company, location, customer segment, or target audience.
Any additional slicer should have a clear analytical purpose and should not make the dashboard unnecessarily crowded.

16.6 Automated Data Refresh

The data preparation process could be extended to support a more automated refresh workflow when the source files are updated and the required connections are available.
This would reduce the need for repeated manual imports.

16.7 Advanced Analytics

Future versions could incorporate statistical analysis, segmentation, or predictive modeling to investigate factors associated with marketing performance.
Such extensions would require suitable data, clearly defined objectives, and appropriate validation.

17. CONCLUSION

The **Social Media Marketing Analytics** project demonstrates the application of Microsoft Power BI to the analysis and visualization of social media and marketing campaign data.

The project uses two datasets to develop separate dashboards for Instagram analytics and marketing campaign analytics.
The Instagram dashboard presents key engagement-related indicators, including posts, likes, comments, shares, and engagement rate. It enables users to explore the available data through content-related and date-based filters.
The Marketing Campaign Analytics dashboard presents important campaign metrics, including impressions, clicks, average ROI, and average conversion rate. It also provides comparisons across campaign types and marketing channels, along with a time-series visualization of impressions and clicks.

DAX measures are used to calculate summary indicators, while charts and slicers provide an interactive reporting environment. Organizing the report into two pages makes it possible to examine different aspects of marketing performance while keeping the datasets analytically separate.

The project demonstrates that data visualization can make large datasets easier to explore and understand. It also highlights the importance of data preparation, correct aggregation, and numerical validation when developing business intelligence reports.
Overall, the project provides practical experience in importing datasets, creating measures, designing dashboards, and presenting marketing information through interactive visualizations.

The final Power BI report serves as a foundation for further work in marketing analytics, including improved data quality checks, additional performance indicators, and more advanced analytical techniques.

18. REFERENCES

The following resources are relevant to the technologies and concepts used in this project.

1. **Kaggle.** Dataset platform used to obtain the source datasets. The exact dataset titles, uploader names, and URLs should be added from the original dataset pages.
2. **Microsoft Power BI Documentation.** Documentation covering data import, data modeling, visualizations, and report development.
3. **Microsoft Learn — Data Analysis Expressions (DAX).** Documentation explaining DAX functions, measures, and calculation behavior.
4. **Microsoft Learn — Power BI Reports.** Documentation covering report pages, visualizations, and interactive filtering.
5. **Microsoft Learn — Power Query.** Documentation covering data import and data transformation.

Reference Links:-

- Kaggle: https://www.kaggle.com/
- Power BI documentation: https://learn.microsoft.com/power-bi/
- DAX documentation: https://learn.microsoft.com/dax/
- Power Query documentation: https://learn.microsoft.com/power-query/

APPENDIX A: DASHBOARD SCREENSHOTS

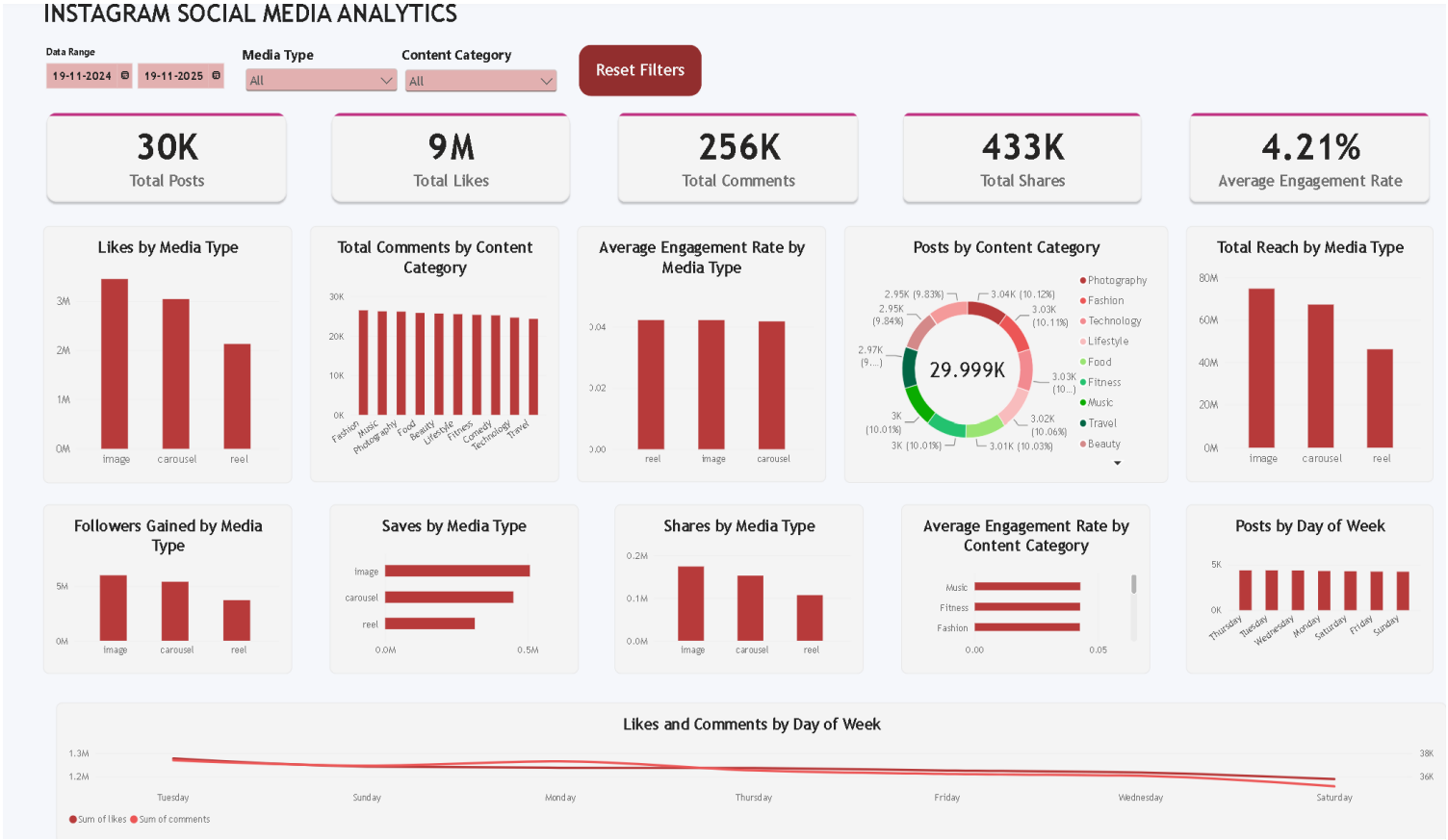


Figure A.1: Dashboard showing Instagram performance indicators, engagement analysis, and interactive filters.

Figure A.2: Marketing Campaign Analytics Dashboard



Figure A.2: Dashboard showing campaign KPIs, campaign-type comparisons, channel performance, and impressions and clicks over time.

APPENDIX B: IMPORTANT MEASURES

The following formulas illustrate the main calculations used in the report. Retain only the formulas that match the final PBIX file.

Total Impressions

```
Total Impressions =
SUM('marketing_campaigns'[Impressions])
```

Total Clicks

```
Total Clicks =
SUM('marketing_campaigns'[Clicks])
```

Average ROI

```
Average ROI =
AVERAGE('marketing_campaigns'[ROI])
```

Average Conversion Rate

```
Average Conversion Rate =
AVERAGE('marketing_campaigns'[Conversion_Rate])
```

Campaign Row Count

```
Campaign Rows Check =
COUNTROWS('marketing_campaigns')
```

Total Posts

```
Total Posts =
DISTINCTCOUNT('Instagram_Analytics'[post_id])
```

Total Likes

```
Total Likes =
SUM('Instagram_Analytics'[likes])
```

Total Comments

```
Total Comments =
SUM('Instagram_Analytics'[comments])
```

Total Shares

```
Total Shares =
SUM('Instagram_Analytics'[shares])
```